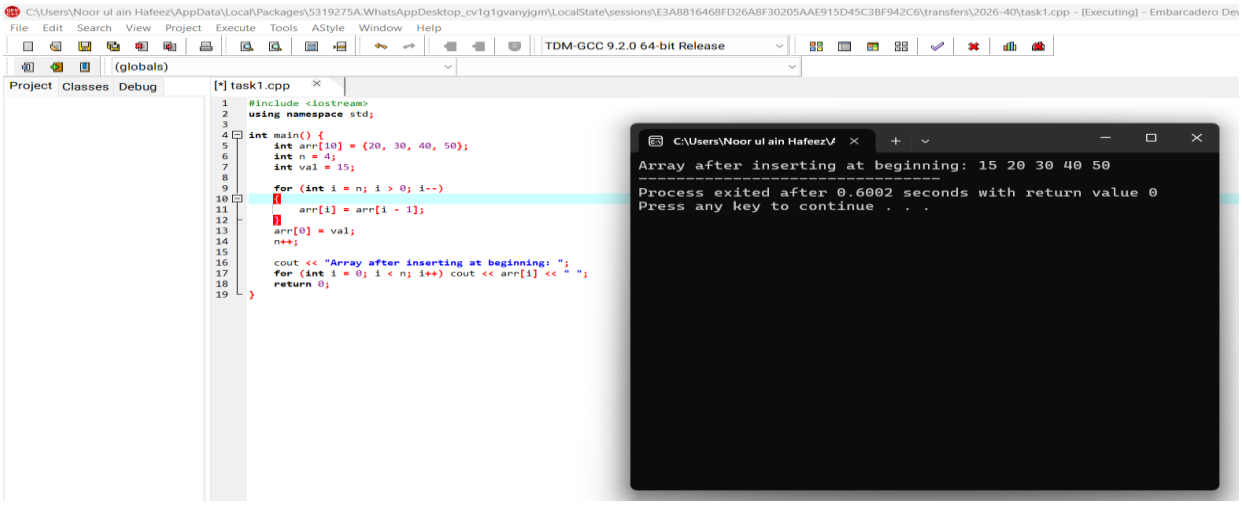
DSA Lab: 4

Submitted By: Noor Ul Ain Hafeez

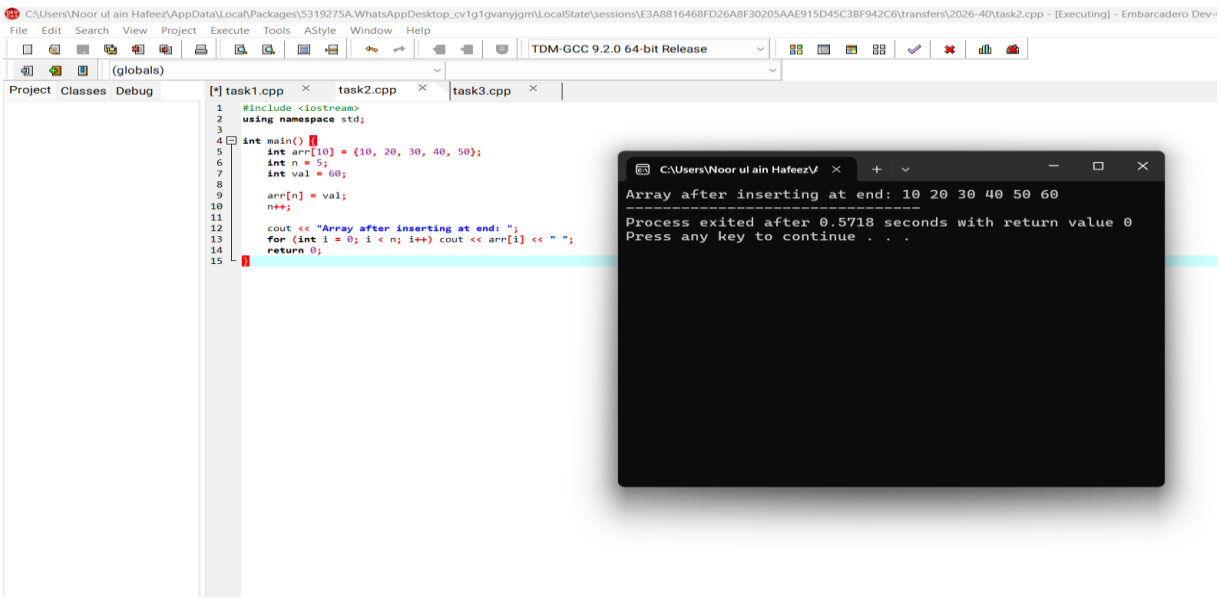
Submitted To: Mam Nayab Khalid

Sapid: 72148

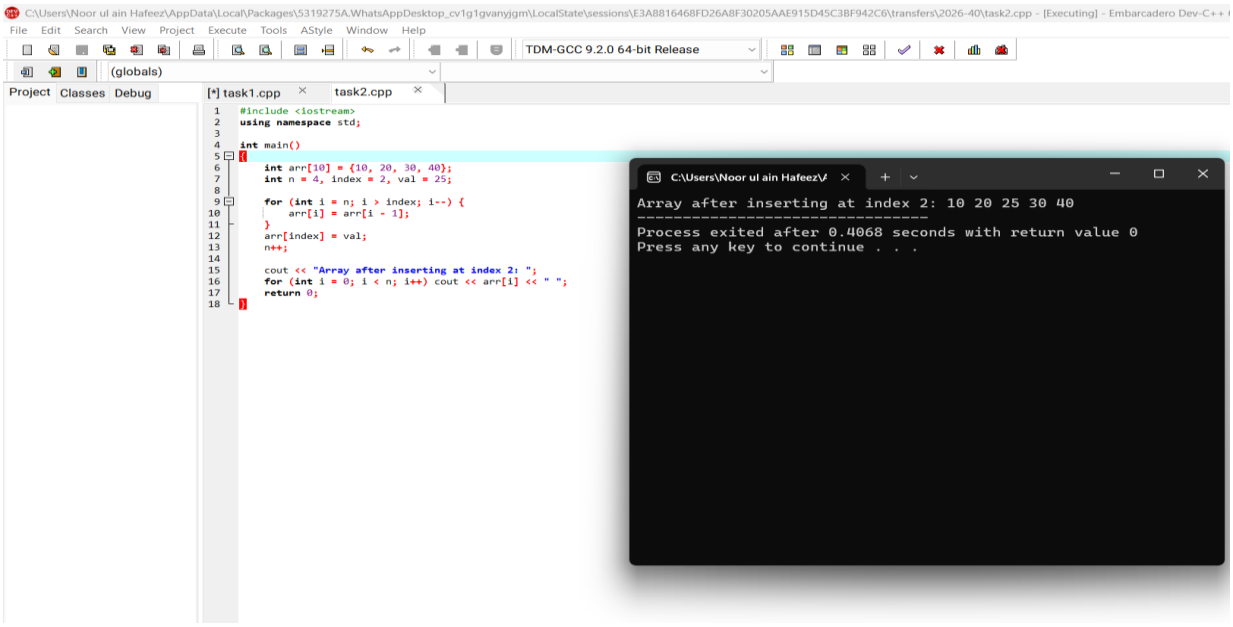
Task 1:



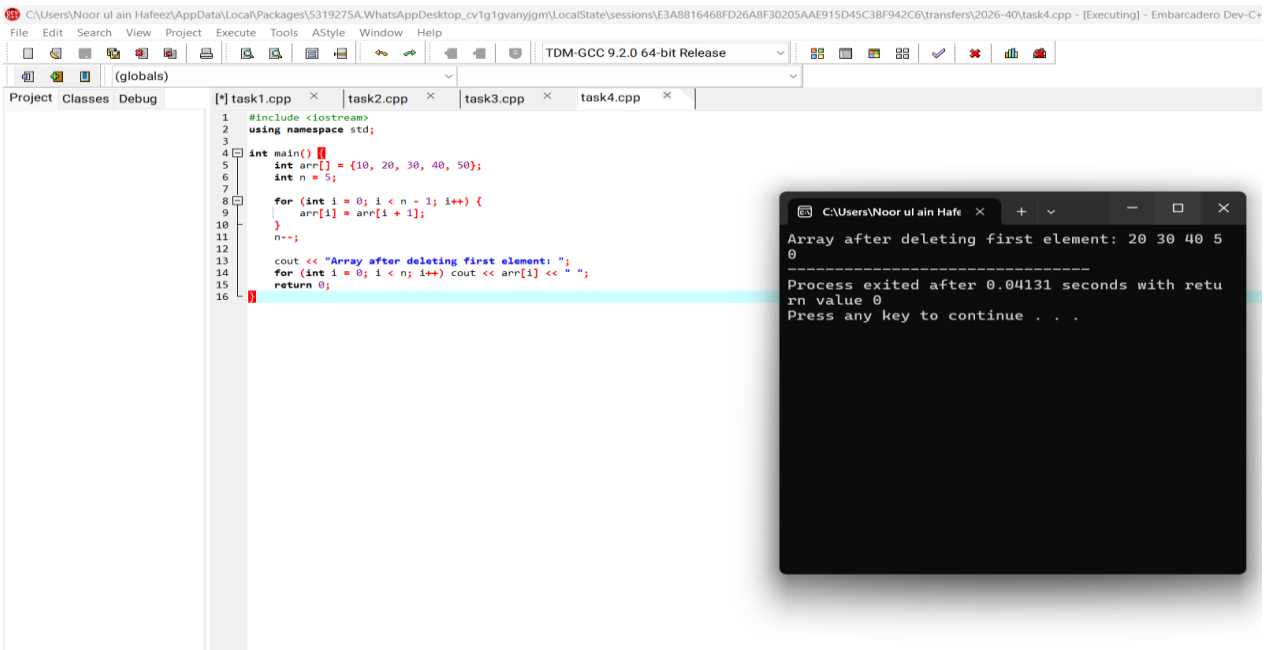
Task 2:



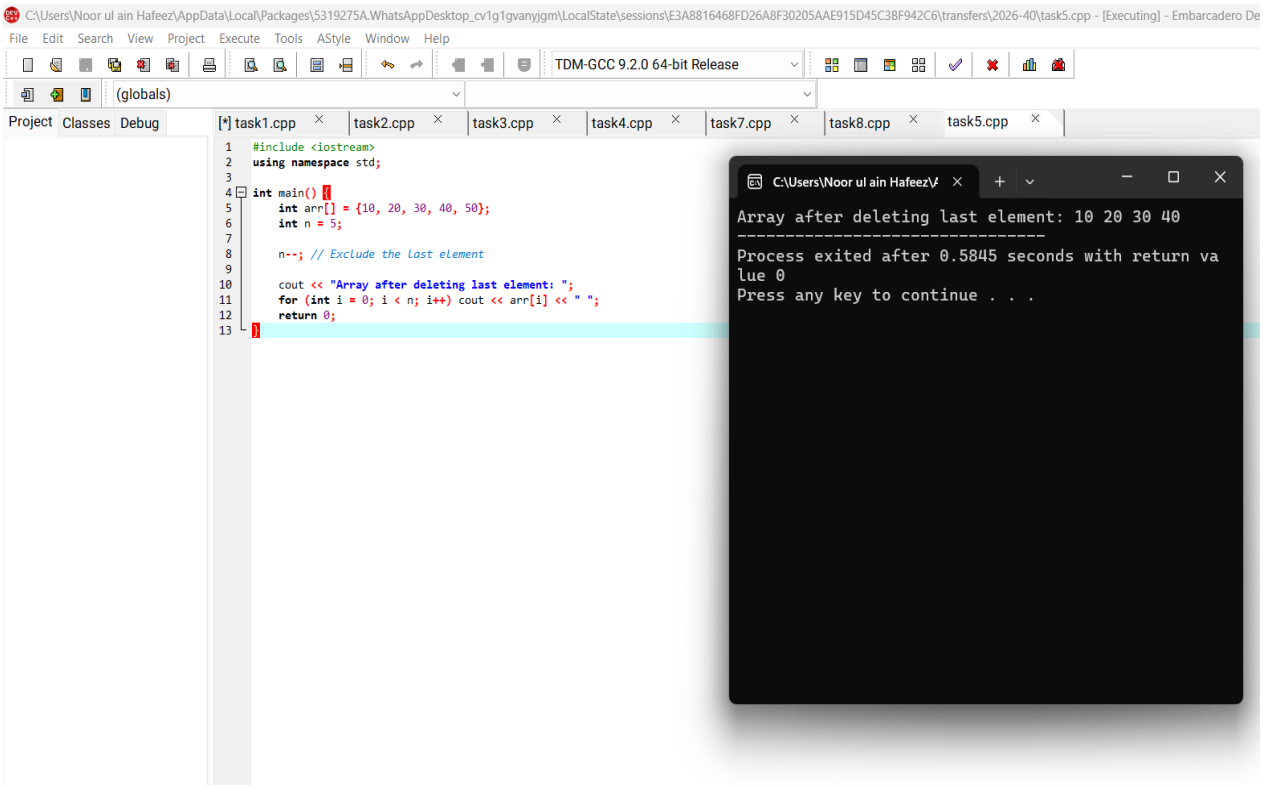
Task 3:


File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 9.2.0 64-bit Release
(globals)
Project Classes Debug
[*] task1.cpp
task2.cpp
#include <iostream>
using namespace std;
int main()
{
int arr[10] = {10, 20, 30, 40};
int n = 4, index = 2, val = 25;
for (int i = n; i > index; i--) {
arr[i] = arr[i - 1];
}
arr[index] = val;
n++;
cout << "Array after inserting at index 2: ";
for (int i = 0; i < n; i++) cout << arr[i] << " ";
return 0;
}
C:\Users\Noor ul ain Hafeez\
Array after inserting at index 2: 10 20 25 30 40
Process exited after 0.4068 seconds with return value 0
Press any key to continue . . .

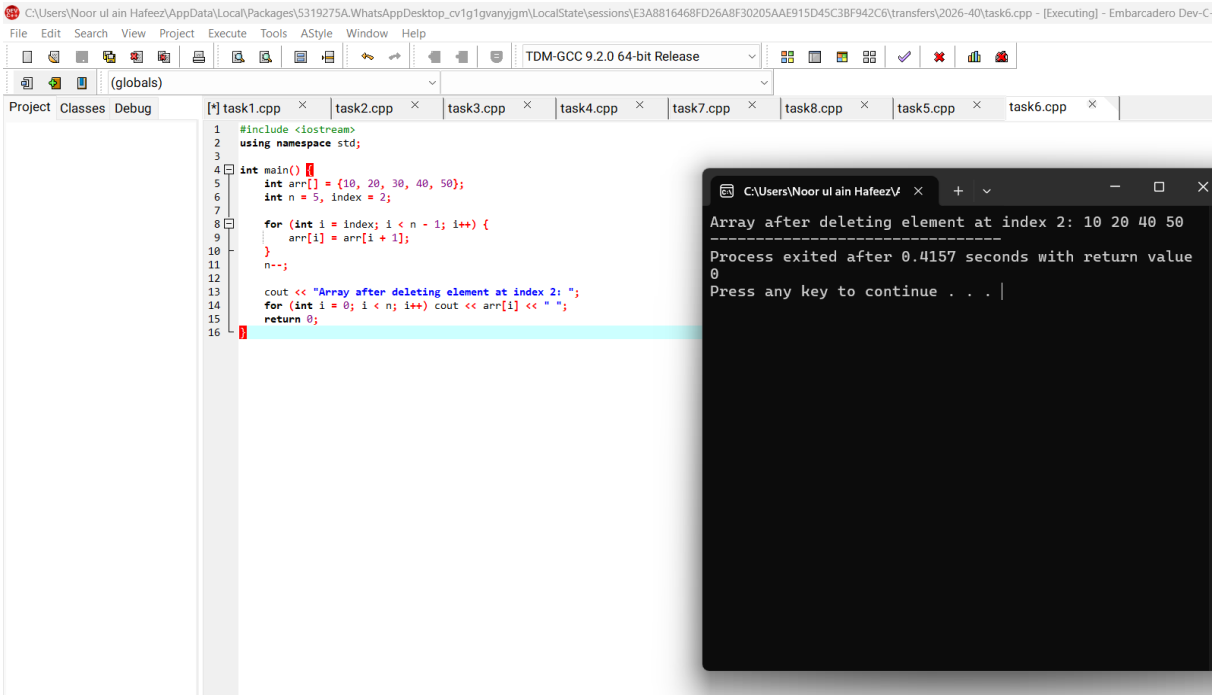
Task 4:


File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 9.2.0 64-bit Release
(globals)
Project Classes Debug
[*] task1.cpp
task2.cpp
task3.cpp
task4.cpp
#include <iostream>
using namespace std;
int main() {
int arr[] = {10, 20, 30, 40, 50};
int n = 5;
for (int i = 0; i < n - 1; i++) {
arr[i] = arr[i + 1];
}
n--;
cout << "Array after deleting first element: ";
for (int i = 0; i < n; i++) cout << arr[i] << " ";
return 0;
}
C:\Users\Noor ul ain Hafe
Array after deleting first element: 20 30 40 5
0
Process exited after 0.04131 seconds with retu
rn value 0
Press any key to continue . . .

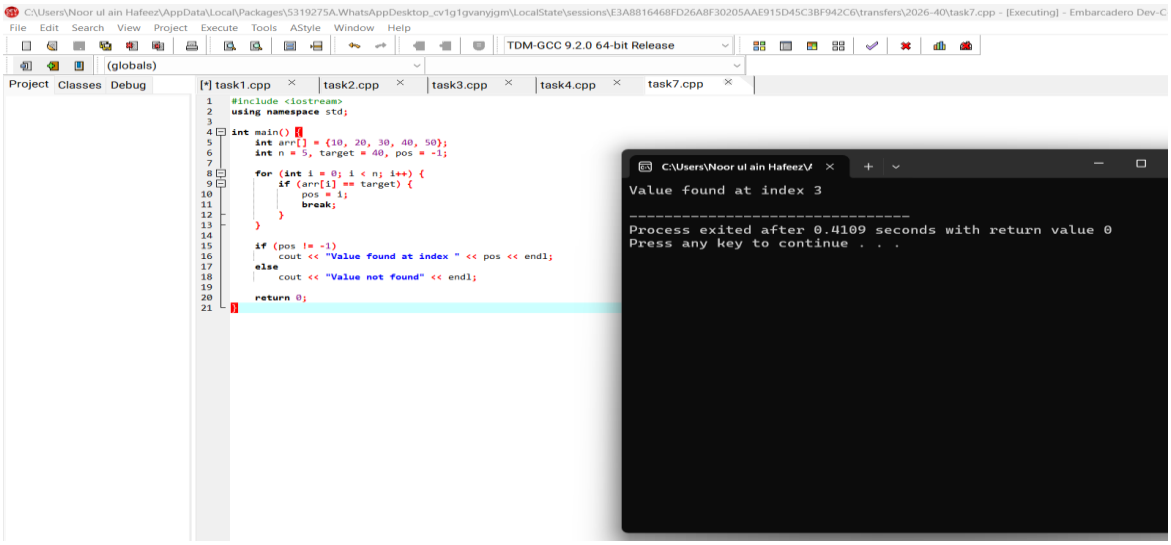
Task 5:


File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 9.2.0 64-bit Release
(globals)
Project Classes Debug
[*] task1.cpp
task2.cpp
task3.cpp
task4.cpp
task7.cpp
task8.cpp
task5.cpp
#include <iostream>
using namespace std;
int main() {
int arr[] = {10, 20, 30, 40, 50};
int n = 5;
n--; // Exclude the last element
cout << "Array after deleting last element: ";
for (int i = 0; i < n; i++) cout << arr[i] << " ";
return 0;
}
C:\Users\Noor ul ain Hafeez\
Array after deleting last element: 10 20 30 40
Process exited after 0.5845 seconds with return va
lue 0
Press any key to continue . . .

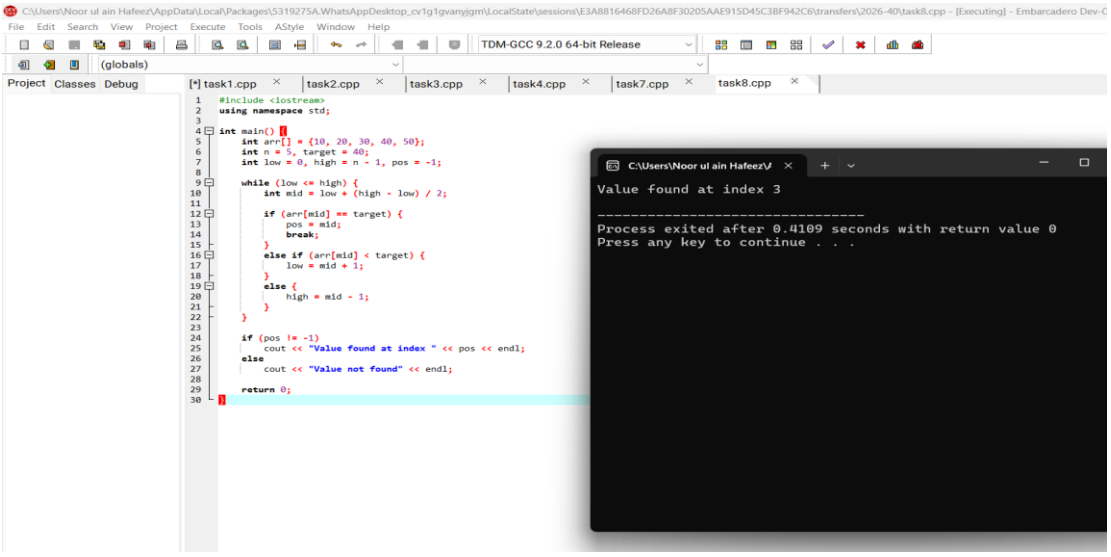
Task 6:


File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 9.2.0 64-bit Release
(globals)
Project Classes Debug
[*] task1.cpp
task2.cpp
task3.cpp
task4.cpp
task7.cpp
task8.cpp
task5.cpp
task6.cpp
#include <iostream>
using namespace std;
int main() {
int arr[] = {10, 20, 30, 40, 50};
int n = 5, index = 2;
for (int i = index; i < n - 1; i++) {
arr[i] = arr[i + 1];
}
n--;
cout << "Array after deleting element at index 2: ";
for (int i = 0; i < n; i++) cout << arr[i] << " ";
return 0;
}
C:\Users\Noor ul ain Hafeez\
Array after deleting element at index 2: 10 20 40 50
Process exited after 0.4157 seconds with return value
0
Press any key to continue . . .

Task 7:


File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 9.2.0 64-bit Release
(globals)
Project Classes Debug
[*] task1.cpp
task2.cpp
task3.cpp
task4.cpp
task7.cpp
#include <iostream>
using namespace std;
int main() {
int arr[] = {10, 20, 30, 40, 50};
int n = 5, target = 40, pos = -1;
for (int i = 0; i < n; i++) {
if (arr[i] == target) {
pos = i;
break;
}
}
if (pos != -1)
cout << "Value found at index " << pos << endl;
else
cout << "Value not found" << endl;
return 0;
}
C:\Users\Noor ul ain Hafeez\A
Value found at index 3
Process exited after 0.4109 seconds with return value 0
Press any key to continue . . .

Task 8:


File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 9.2.0 64-bit Release
(globals)
Project Classes Debug
[*] task1.cpp
task2.cpp
task3.cpp
task4.cpp
task7.cpp
task8.cpp
#include <iostream>
using namespace std;
int main() {
int arr[] = {10, 20, 30, 40, 50};
int n = 5, target = 40;
int low = 0, high = n - 1, pos = -1;
while (low <= high) {
int mid = low + (high - low) / 2;
if (arr[mid] == target) {
pos = mid;
break;
}
else if (arr[mid] < target) {
low = mid + 1;
}
else {
high = mid - 1;
}
}
if (pos != -1)
cout << "Value found at index " << pos << endl;
else
cout << "Value not found" << endl;
return 0;
}
C:\Users\Noor ul ain Hafeez\A
Value found at index 3
Process exited after 0.4109 seconds with return value 0
Press any key to continue . . .